



Stated and revealed exit choices of pedestrian crowd evacuees



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ABSTRACT

Introduction: Understanding fundamental behavioural features regulating the escape wayfinding decisions of pedestrian humans in built environments has major implications for the safety planning and the risk-analysis of crowded public facilities. In contrast to the vast interest invested in modelling the momentary responses of evacuees to their immediate surroundings (i.e. the collision-avoidance models), their global navigation behaviour is poorly understood albeit believed to be a major determinant of the accuracy of the crowd simulation models. The primary challenge arises from the scarcity of reliable data suitable for modelling purposes causing the experimental knowledge in the field lagging substantially behind the corresponding model developments. Observations derived from fully natural emergency contexts (in the form of modelling material) are rare and collecting data in realistic experimental settings poses its own major challenges. Only few experimental modelling attempts have been reported thus far in relation with this problem primarily using the stated-choice (SC) methods. Modelling based on revealed choices (RC), however, has remained absent in this context leaving the findings of the SC experiments mostly unverified. It is still unclear whether we can reliably learn from the wayfinding choices made in virtually visualised environments without the decision-makers interacting with real individuals and the physical elements of the environment as they do in the real-world settings. Furthermore, the extent to which the findings of these experiments are specific to the particular characteristics of the environment visualised in the experiments is also unclear.

Methods: To bridge this gap, here we report on discrete-choice estimates derived from observations of both types. Three datasets of stated exit choices (4958 observations) were collected through face-to-face interviews with pedestrians in three public places with the experiments referring to the particular geometry of the place in which the participants were interviewed. Also, 3015 disaggregate real (or more precisely, “realistic”) exit choices were extracted through individual-level video analysis of the footage of a series of novel evacuation trials that simulate pedestrians’ emergency escape. The participants competed and interacted with real individuals in an actual crowd and made actual wayfinding decisions to make the quickest possible escape. Our particular interest was investigating the sample-to-sample variations and the context-dependence of the inferred estimates.

Results: We observed fairly similar parameter estimate patterns emerged from all four datasets, and identified many behavioural aspects upon which all models consistently agreed regardless of their context of origin. Moreover, despite the significant differences

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between the parameter scales of the four models, the SC-generated models made predictions that were reasonably similar to those of the RC model as well as to those of the model derived from the combined data.

Applications: Our findings provide promising evidence as to the potential applicability of the SC methods in particular as well as other forms of virtual-reality decision experiments in general as a practical, flexible and ethical approach for the continuation of research and advancing the state of knowledge in this field.

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1. Introduction

The growing urban populations as well as the increasing occurrence of public events have given rise to a rapid rise in the frequency and the scale of congregations of large numbers of pedestrian humans in public spaces. Recent disastrous incidents have shown that crowded places can be exposed to significant safety dangers and that the presence of a large number of pedestrians can potentially result in injuries and fatalities at large scales if not planned and managed reasonably (Helbing et al., 2002; Helbing et al., 2007; Still, 2014). This has raised concerns as to the safety of crowded public facilities and has led to a growing body of research in various disciplines attempting to understand different aspects of human crowds and predict their behaviour in different situations and scenarios. Enhancing the state of knowledge about humans' behaviour in crowded spaces (in both emergency and non-emergency situations) is a necessary step towards the development of reliable modelling tools (Cheung and Lam, 1998; Daamen et al., 2005; Do et al., 2016; Kneidl and Borrmann, 2011; Kneidl et al., 2013). It will enable planners to perform reliable risk-analyses on the design and performance of their facilities, thereby improving the occupants' safety through taking effective planning or design measures.

There are several aspects of the problem that pose serious modelling challenges to the researchers. Among those is the context dependence of the emergent behaviour and the variety of contexts with which the modeller need to deal. Crowds can form at different sizes and for different purposes and activities such as sport events, political protests, religious gatherings, or they could simply form as a result of the rush of demand to public transport facilities and public buildings in peak hours. Also, from a spatial point of view, there is a large variety of geometric types in which crowds can be formed ranging from confined rooms to large open spaces, and to complex built environments or high-rise buildings. Clearly, each of these aspects and their various possible combinations can significantly impact on how the crowd may behave and thus necessitates dedicated modelling and empirical observations. This has made prescription of a universal crowd model that embodies all these aspects extremely challenging, if not impossible.

One particular topic that has received an increasing attention in the recent years is the escape and evacuations of pedestrian crowds when emergencies arise in complex built environments (Guo et al., 2011). The problem is of crucial importance and has major safety-related implications. However, realistic observations obtained from the real-world emergency contexts that can serve as modelling material are extremely rare (Haghani et al., 2014; Haghani and Sarvi, 2016; Shahhoseini et al., 2016). On the other hand, owing to the intrinsic complexity and diversity of the human behaviour as well as the particular behavioural features that might arise during emergencies, pure modelling intuition is not likely to provide a sufficient answer to the problem, and this highlights the importance of empirical experimentation in this context.

Here, we focus on escape behaviour of pedestrian humans when emergencies arise in complex confined spaces. We mainly focus on situations where evacuees are subject to non-trivial navigational decisions stemmed from the sudden rush of the crowd and the drop of capacities at exit route options. There have been a number of studies attempted to model this problem based on empirical evidence among which stated-choice (SC) methods (Duives and Mahmassani, 2012; Haghani and Sarvi, 2016a; Haghani et al., 2015a; Haghani et al., 2015b; Lovreglio et al., 2014) or experimentations in virtual decision environments (Bode et al., 2014; Bode and Codling, 2013; Kinatader et al., 2014a; Kinatader et al., 2014b; Moussaïd et al., 2016) have proved as a popular and practical approach. In this class of studies, participants have been introduced to virtually visualised scenarios of emergency escapes from crowded places and have asked to make wayfinding (e.g. exit) decisions. However, taking into account the particular contextual elements presented by the real evacuations that are mostly failed to be represented in SC settings, major concerns can be expressed as to the reliability of findings obtained from the SC experiments in this context. This has left the applicability of the SC-generated exit-choice models unverified and subject to scepticism thus far. The absence of the revealed-choice (RC) observations, through which this question can be objectively investigated, has remained as a major research gap in this context despite the clear implications of this problem for the continuation of the research in this area. Moreover, it is not clear the extent to which the findings emerged from these SC experiments are specific to the particular geometrical features of the environments visualised in these experiments.

Here, we intend to address this problem by providing choice observations of both kinds. We conducted experiments of stated emergency exit choice by interviewing participants in public places as they left those places. As opposed to the common approach of the internet-based surveys, our choice experiments make partial references to the individuals' recent real-life decision merely on the account of the particular geometry visualised in the choice scenarios. We conducted this

type of experiments in three public places (two urban train stations, and one public building in Melbourne) and interviewed 400 individuals resulting in a total of 5635 observations of stated exit choices. In addition, we report on a series of realistic field-type trials of mock evacuations¹ through which we extracted revealed or real (or more precisely, “realistic”) experimental observations of emergency exit choice. Our participants (150 individuals) performed the simulated evacuation tasks by competing with one another to escape from certain artificially-built multi-exit spaces as quickly as possible. Their movements’ trajectories and behaviour were closely analysed on an individual-level basis through which 3015 RC observations were extracted.

Discrete-choice mixed and latent-class logit estimates were inferred from the observations obtained from these four choice datasets. Of particular interest is to understand how the resultant modelling findings (the inferred estimated as well as the models’ predictions) vary from sample to sample and also how the change of context (e.g. SC versus RC context, or different SC contexts with reference to different survey locations) impacts on the estimates and the predictions. This will also enable us to identify certain aspects that are generally agreed upon by the majority of the observations and exclude the aspects that are more likely to be specific to a particular sample or experimental context. In doing so, we merely put the estimates and the predictions of the models into comparison. From this perspective, this study is not particularly aimed at identifying the model that results “the best” or “most accurate” predictions or the best modelling fit, nor is the quantitative extent of hypothetical bias (Beck et al., 2016; Hensher, 2010) (i.e. the absolute deviations of the SC-generated estimates (or their rations) from their RC-generated counterparts) of particular interest of this work. We suggest that addressing this issue may require dedicated analyses. Nonetheless, considering the model obtained from the RC observations as the benchmark (or the more reliable set of observations), one may be able to acquire insights from our analyses as to the accuracy of the predictions or estimates of each single SC dataset.

2. Background

In the framework of the individual-based (i.e. microscopic) modelling of crowd evacuations (Guo and Huang, 2012), three major levels of decisions are recognised to be made (and as a result, need to be modelled) by each individual pedestrian evacuee. The categorisation and the respective terminologies have proved useful in providing a common language for communicating in the context of crowd modelling and has received growing acceptance in the literature in the recent years (Asano et al., 2010; Haghani and Sarvi, 2016b; Haghani et al., 2015b; Hoogendoorn and Bovy, 2004). At the highest level of the hierarchy, decisions need to be made by each individual for initiating the evacuation. This is the decision that defines the “pre-evacuation time” and is often referred to as the “strategic decision” (Lovreglio et al., 2015). Then the evacuee sets a goal or target that leads them outside of the surrounding environment. This is the decision that defines the “route and exit choice” and is often referred to in the literature as the “tactical decision” (Bode and Codling, 2013). Finally, in order to reach the target (e.g. exit) decided at the previous stage, the pedestrian needs to make momentary decisions to prevent collision with obstacles and other pedestrians while moving towards the target. This is the decision that defines pedestrians’ “next step” and is referred to as “operational decisions” (Antonini et al., 2006). Interestingly, this categorisation is to a great extent similar to the hierarchical control model that explains how the sensorimotor system in humans’ neural system performs when generating voluntary movements (Loeb et al., 1999).

To our knowledge, a great deal of research has been dedicated in the existing literature towards understanding and modelling the next-step (i.e. operational-level) choices of pedestrians which has led to the development of fairly robust walking behaviour models (Moussaïd et al., 2011; Robin et al., 2009; Wang et al., 2015a; Wang et al., 2015b). However, much fewer modelling attempts have been made towards understanding the two other decision types and as we go upwards in the previously-mentioned hierarchy of the evacuation decisions, less is known about the nature of the problem (Huang and Guo, 2008). Clearly, reliable modelling tools of crowd evacuation necessitate consistency in how accurately these three decision types are replicated in the crowd simulation methods.

Among the earlier studies that have modelled the problem of evacuees’ tactical choices, two common data provision approaches can be generally identified. This includes experiments in virtual-choice settings (e.g. SC experiments) and the field-type trials that involve movement actions of real humans in controlled experimental settings. In a number of interrelated articles Bode and Codling (2013), Bode et al. (2014) and Bode et al. (2015) reported on performing laboratory-based computer experiments that address pedestrian route choice problem. The experiments require participants to interact with other simulated agents (i.e. virtual evacuees) by clicking on a computer mouse and leading the single subject that they have the control of to their chosen exit while competing with simulated entities that make up the virtual crowd. The experiments

¹ As pertinently pointed out by a referee of this paper, we would like to acknowledge the fact that both of the observation types reported in this work are in fact of experimental (i.e. laboratory) nature in strict terms. The survey-based experiments are based on purely hypothetical scenarios, whereas the other experiment is more realistic in that people do actually compete with other individuals in real crowds to escape. The use of the terms “realistic” or “field” should not be however misinterpreted as if the observations are claimed to have been collected from fully-natural settings. Yet, it is not uncommon (particularly in the literature of experimental economics) to refer to such experiments in which the interference of the experimenters are limited to minimum as “field-type lab experiments” (Viceisza, 2015) or simply, “field-type” experiments as opposed to the fully hypothetical experimental environments in the same context. In other words, experimental economists recognize a range of field experiments (see Harrison and List, 2004) for a detailed taxonomy, and here we borrow the same type of terminology just for the sake of simplifying the communication throughout the paper.

were executed under different types of treatment such as “time-independent”, “time-dependent” and “memorised information”. They also introduced certain elements of competitiveness using verbal motivation (in fact, written messages appearing on the screen) as proxies of “time pressure” or “stress” (referred to in their article as “stress-induced” treatments). Their experiment did not generate individual-level choice data in its classic definition (i.e. with the components of choice-set, attribute levels and dependent variable), nevertheless their statistical analysis provided useful findings into how individuals make exit route choices in a crowded setting. For example, they suggested that their subjects did not show a strong tendency for choosing familiar routes and that stress-induced treatments adversely impacted on the optimality of their decisions. Also “dynamic information” proved more effective in influencing their subjects’ decisions than “static information”. In other studies, [Lovreglio et al. \(2014\)](#) and [Duives and Mahmassani \(2012\)](#) reported on similar types of internet-based experiments but in the form of SC methods. The former investigates the impact of demographic characteristics of individuals on exit decision making while the latter mainly focuses on the impact of the implementation of their utility-based exit-choice model to the simulation phase and contrasting it with the common predefined behavioural rules such as the shortest path decision criterion. These experiments share common characteristics in terms of the experiment design that includes application of the binary-choice scenarios, having been performed in the form of internet-based choice surveys and the simplicity of the visualised geometries (relative to the complex architectural designs that may typically be found in real public places).

An alternative to the computer-based experimentations is field-type trials that involve real humans making real movements and making actual choices in controlled but more realistic environments relative to the virtual-decision experiments. This method, however, offers less flexibility and control over variations to the analyst but is regarded to be more reliable as the participants have a more realistic perception of their surrounding environment as well as their interactions with other individuals. We know of three experiments of this type addressing the particular problem of evacuees’ tactical choices. [Guo et al. \(2012\)](#) contrasted the route finding behaviour of evacuees in zero visibility conditions with the condition of intact vision through experimentations in a classroom geometry. [Heliövaara et al. \(2012\)](#) conducted an experiment in a corridor with an asymmetrical exit layout where their key finding suggested that evacuations might take longer when individuals try to behave cooperatively compared to when they behave egoistically (i.e. trying to minimise their own evacuation time). Two other studies conducted by [Liu et al. \(2009\)](#) and [Fang et al. \(2010\)](#) primarily focused on making verification efforts for their proposed modified Cellular Automata simulation method using. The experiment by [Liu et al. \(2009\)](#) was conducted in a classroom with two exits and showed that occupants’ density around the exits might as well play an important role in crowd evacuation in addition to the spatial distances. On the contrary, [Fang et al. \(2010\)](#) found a strong tendency of evacuees simply choosing the closest exit in their experiment especially when the crowding level is small or at medium level. Also, a common finding suggested by majority of these studies is the sub-optimality of the pedestrians’ route choices in a crowd in terms of minimising their egress time. There are also common characteristics shared by these experimental studies that pose some limitations towards the reliability of their emerged findings. They have mostly been conducted in bi-exit settings, with the exits being located often too close (almost adjacent), a restrictive geometric feature that hinders the identification of many contributing factors to wayfinding decisions. They also have mostly taken place in calm normal-walking situations which does not strongly represent the contextual elements of emergency scenarios in that little or no competition is induced by the experiment settings for making the quickest exit. Most importantly, though, the information extracted from those observations has merely concerned the aggregate (i.e. macro-level) measures of behaviour such as the “egress shares” of exit alternatives. Although great insight has been obtained from such observations, the level of information extraction in these experiments has barely gone beyond the macro-scale measures such as flow, density and total egress time. Choice observations at the disaggregate level that could serve as modelling material have particularly not been provided by the former studies.

3. Experimental methods and materials

3.1. SC experiments

The SC experiments were conducted in the form of face-to-face interviews with pedestrians leaving certain public facilities in Melbourne, Australia. Choice scenarios were presented to the participants as pictures that visualised the choice situations in simplified maps of the same place which the pedestrian had just exited at the time of experiment. In this sense, the experiments made partial references to the real-life experience of the decision makers, restricting the reference merely to the geometry depicted in the choice scenarios. The survey locations included a public building (Robert Menzies building) and two major urban train stations in Melbourne (the Flinders-street and Parliament station, with the latter being an underground station). We respectively interviewed 168, 122 and 120 individuals in these three locations (410 individuals in total). Here, we refer to these three experiments as SC (I), SC (II) and SC (III) respectively. [Figs. 1–3](#) respectively show examples of the choice situations introduced to the participants in each of these experimental settings. The participants were asked to hypothesise each visualised scenario as a decision situation in an emergency evacuation context and were asked to choose the alternative exit that would choose if they faced such situation in the real life.

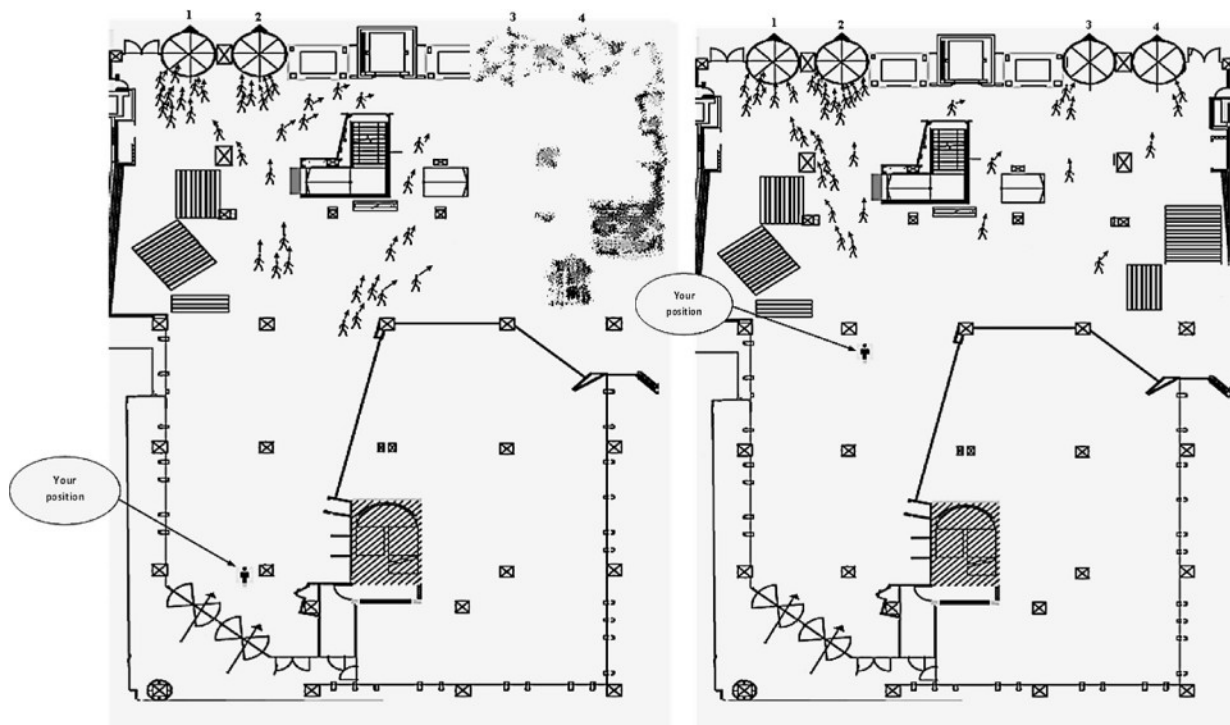


Fig. 1. Samples of the SC (I) choice experiments.

Each choice situation is a unique combination of four design variables. This includes (1) the spatial *distance* of the decision maker to each exit (DIST) presented to the subject by showing their hypothetical position in the picture, (2) the level of *congestion* (CONG) of occupants around each exit door (i.e. the number of pedestrian evacuees congregating around each exit), (3) the *size of directional flows* of pedestrians (i.e. the number of evacuees) moving towards each exit (FLTOEX), and (4) whether or not the exit is *visible* to the subject (VIS), dummy coded as a variable that takes the value of 1 if the exit is visible and takes 0 otherwise. The distances shown on each picture were measured in the actual locations of the surveys. The first author would position himself at the approximate locations shown on each picture and traverse the distances to each available exit at that venue while measuring the distance using a surveyor's wheel. The measurements were averaged over a few repetitions for each case. The invisibility condition for each exit was assumed to be the result of the hypothetical position of the decision maker inside the visualised geometry. When the visibility of a certain exit was blocked due to the architectural features of the environment, the area around that particular exit was blurred in the picture and the CONG variable associated to that particular exit was accordingly not shown to the decision maker. We refer to this effect as “partial attribute ambiguity”. In all cases, however, the subject had the option to choose any of the available exits regardless of their visibility and that includes the alternatives whose full attributes were given to them as well as the ones with partially ambiguous attributes.

The choice set sizes in the three SC experiments were respectively 4, 6 and 3. The design for the SC (I) and SC (II) experiments consisted of 2 and 3 blocks respectively, while we designed only one block of scenarios for SC (III) experiment. For instance, we introduced customised blocks of experiments according to the exit alternative that the participant had chosen in the real-life (normal) exit setting.² For example, in SC (II), the block of experiments for the individuals who were interviewed in front of the Exits A, B and C (as their chosen exit in real-life situations) were different to those of the interviews conducted in front of the exits E and F, or Exit D. The use of multiple experiment blocks was primarily practiced to create more variability in the SC observations datasets essential for making efficient estimates (Shahhoseini et al., 2015). Also, in some cases (particularly in the survey conducted at the Flinders-Street station) it helped the experimenters to create hypothetical situations that resembled more closely to the choice situation that the individual had just experienced in real life under the normal (i.e. non-emergency) condition. For example, if the individual was observed to have chosen Exit A in their non-emergency choice of exit conducted at the Flinders-Street station (the SC (II) survey), they would then

² We would like to emphasize here that we did not record any RC observation from this setting of experiments, as there was no emergency at those places at the time of the experiments and we also did not record any attribute level for their real-life choices. We only observed the exit that they chose in the real life situation as the interviews were in fact conducted at the exit points.

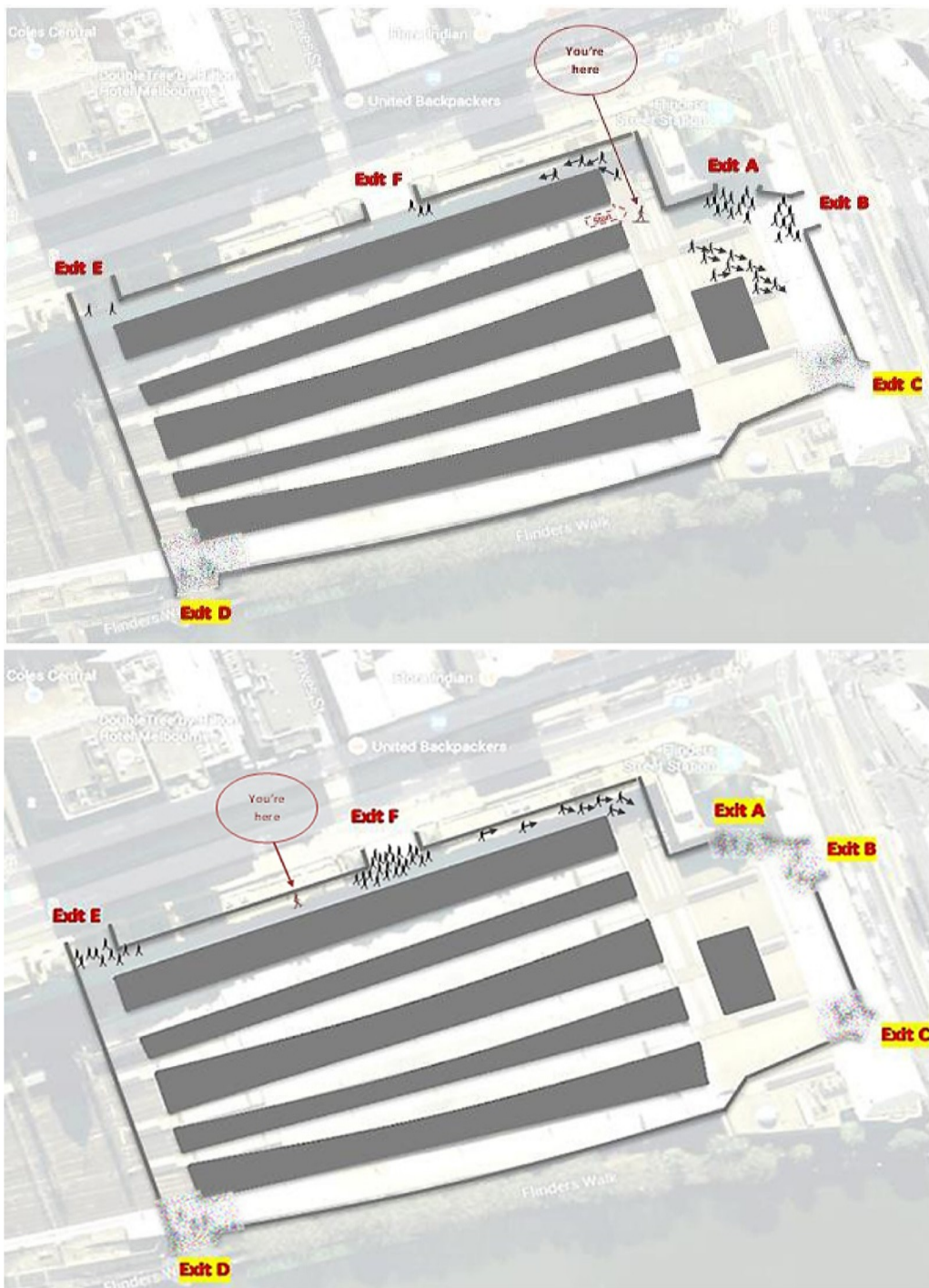


Fig. 2. Samples of the SC (II) choice experiments.

subsequently introduced to a block of experiments in which the (hypothetical) choice maker is close to Exit A but Exit A was mostly inferior compared to other exits in terms of the level of congestion. The number of choice situations ranged from 10 to 14 over the different blocks and in total, 57 unique choice scenarios were designed. The number of choice observations extracted from the three experiments was 2338, 1410 and 1210 respectively (4958 observations in total). Details of the scenarios introduced in each block of these three experiments can be found in the supplementary materials published online alongside this article (Supplementary Methods).

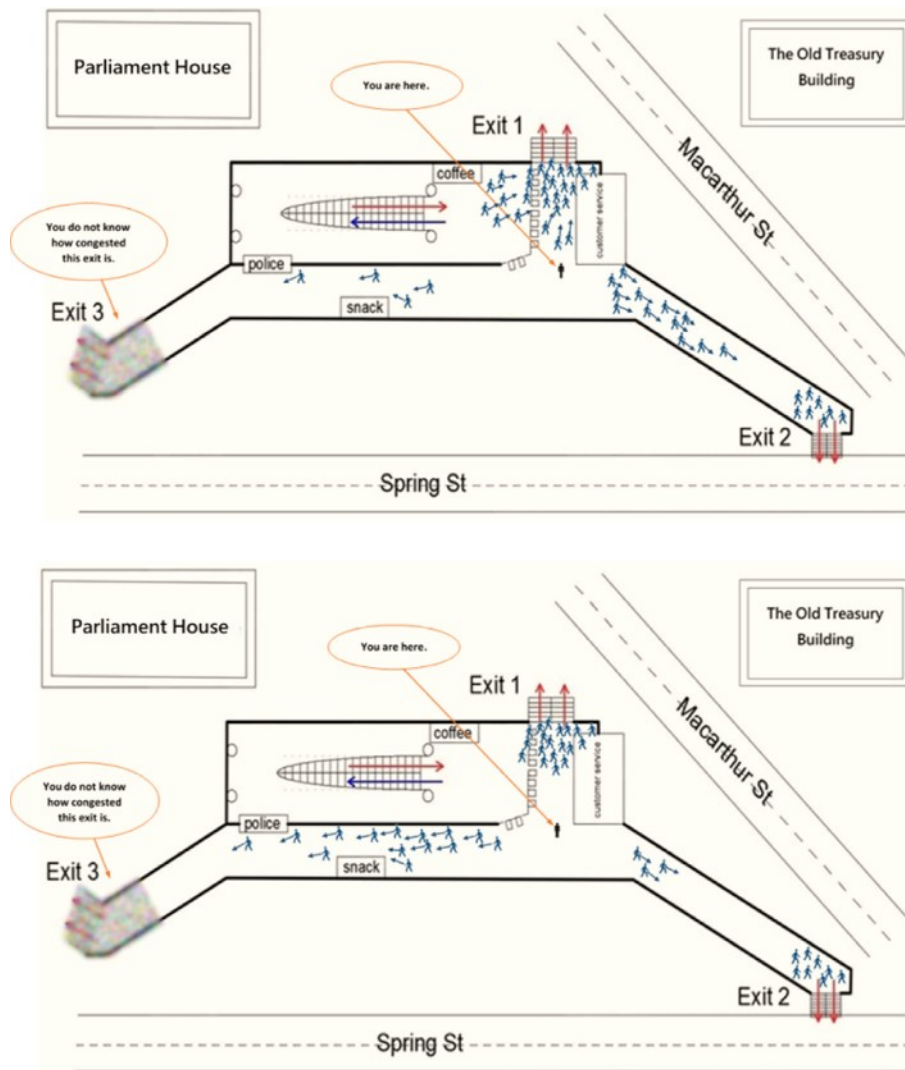


Fig. 3. Samples of the SC (III) choice experiments.

3.2. RC (field-type) experiments

The field-type trials from which we extracted realistic (or revealed) observations³ of emergency exit choice were conducted on 25th of February 2015 at a sport centre in Melbourne, Australia. One hundred and fifty participants were recruited to perform the simulated evacuation tasks. The trials simulated the escape decision making of humans in multi-exit confined spaces that the experimenters temporarily built for the purpose of these trials. The general geometry of the evacuation room was inspired by the design of the Robert Menzies building which was the geometry reference of our SC (I) experiments, although the artificial rooms of experiments had to be built at a small scale than the actual location. We also made certain variations to the distribution and the number of exits over the trial runs compared to the actual design of the building. The experiment was comprised of 20 runs in total. Each run was a unique combination of certain design factors including the number of exits (2, 3 or 4), the number of evacuees (75 or 150), exit widths (either 50 cm or 100 cm) and the spatial distribution of exits. Fig. 4 illustrates the mid-frame snapshot of the raw footage of four trial runs. Full details of the design factors at each trial run, as well as the mid-frame picture of each run can be found in the supplementary material of this

³ The authors would like to acknowledge the fact that the observations obtained from the realistic experiments are themselves of laboratory nature and the use of term “revealed” may be perceived arguable here from this perspective. Considering that choice data extracted from fully naturally-occurring settings of evacuation (i.e. the genuine revealed choice in its classic definition) is absolutely rare, we refer to these observations as “revealed” or “real” only relative to the picture-based SC experiments and for the sake of simplicity. Otherwise, we do acknowledge that “realistic” is a more accurate description of these observations.



Fig. 4. Snapshots from the raw footage of the field-type trials. The participants wait in the initial position located on the top right of the figures and start their escape by entering the room and evacuating it through the available exit alternatives. The quantity, position and capacity of the exits vary from run to run.

article (Supplementary Methods). By making variations to the combination of these design factors we intended to create a great range of variability of the type of choice scenarios that subjects would face over the trial runs. This also helped us prevent the effect of habituation to certain degrees as the participants were not aware of the exact configuration of the room (i.e. the quantity, position and capacity of the exits) prior to each run (although they knew that they will be able to find exits at both sides of the obstacles inside the room). At each run of the trials, participants were instructed to compete with one another to run away from the start point of the evacuation room as quickly as possible and leave the confined space using the available exit options hypothesising that they were escaping from an acute danger in that environment.

The trials were recorded from a top-down view using a camera mounted at 8 m height from the floor of the experiment location. We had our participants to wear green colour beanies in order to recognise their position and track their movement. Their movement trajectories were extracted using the colour-recognition mode of the PeTrack software (Boltes and Seyfried, 2013). We analysed the movements and the trajectories of the participants individually as they moved towards the exits. By doing so, we identified the moments at which they most likely made their exit decision. We used their head orientation and body movement in addition to their trajectory as external clues for their decision moments. In general, we defined the decision moment as the point in their movement after which they did not significantly explore different alternatives and showed least degree of uncertainty about their decision. In Fig. 5, we have illustrated this by providing the snapshots of four individuals at their identified decision moments as they appeared in the PeTrack analysis environment along with their full movement trajectories. Individuals for whom multiple decision moments were identifiable contributed more than one observation to our dataset. We observed individuals who made a clear initial decision, but as they progressed towards their chosen exit they changed their mind and shifted their decision to another alternative. We exemplified this in Fig. 6 by showing one of these individuals at their initial and the subsequent updated decision moment.

At each identified decision moment we extracted all the choice-relevant information and recorded them as a choice observation. This includes the number of alternatives (i.e. the choice set size) at that particular trial run, the choice (i.e. the chosen alternative) and the exit attributes. The exit attributes that we recorded were precisely the same as the attributes used to generate the SC experiments. As exemplified in Fig. 7, for each observation, we recorded the spatial distance of subject to each exit (i.e. the distances, DIST), the number of evacuees congesting around each exit (i.e. the congestions, CONG), the number of evacuees moving ahead of the subject towards each exit (i.e. the size of directional flows to the exits, FLTOEX); and the visibility condition of each exit from the position at which the subject made their decision (the dummy-

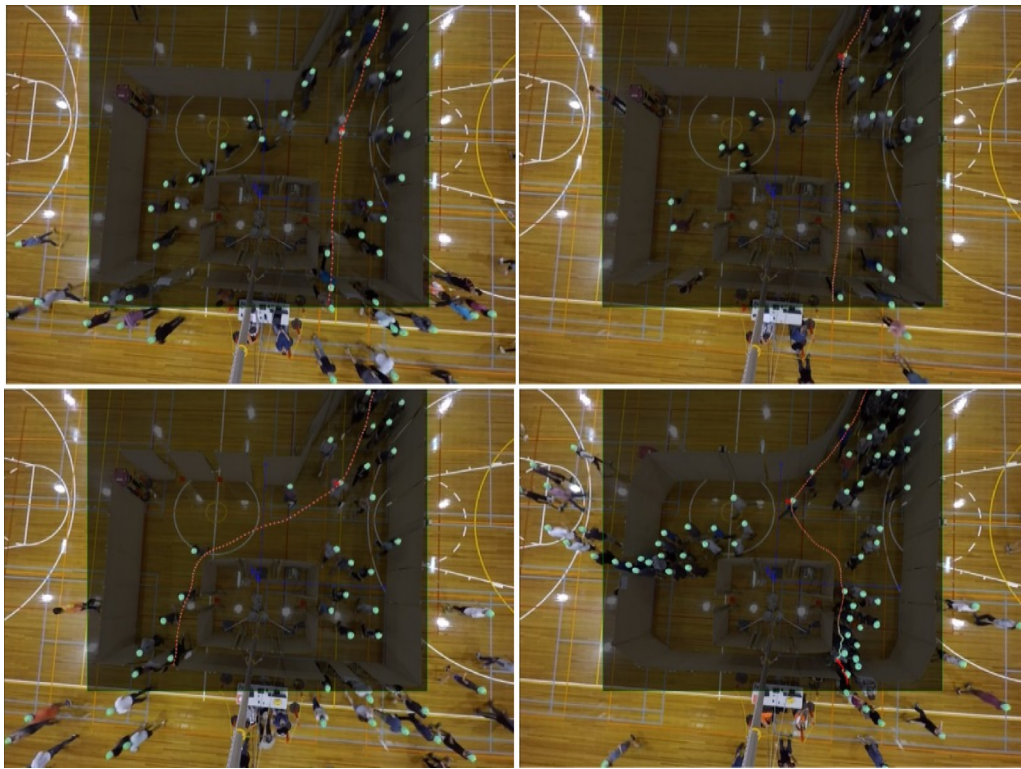


Fig. 5. Examples of four participants at their identified decision moments along with the full trajectory of their movement as they appeared in the trajectory extraction software environment. The body movement and head orientation of the individuals are examined as they moved and as their trajectory was extracted. Their decision moment is defined as the moment after which they showed the most level of consistency in their movement and effectively stopped exploring different alternatives (reflected mainly in their head orientation). Abrupt changes in the trajectories were often followed by consistency in their movement and were considered as major clues for identifying the decision moments.

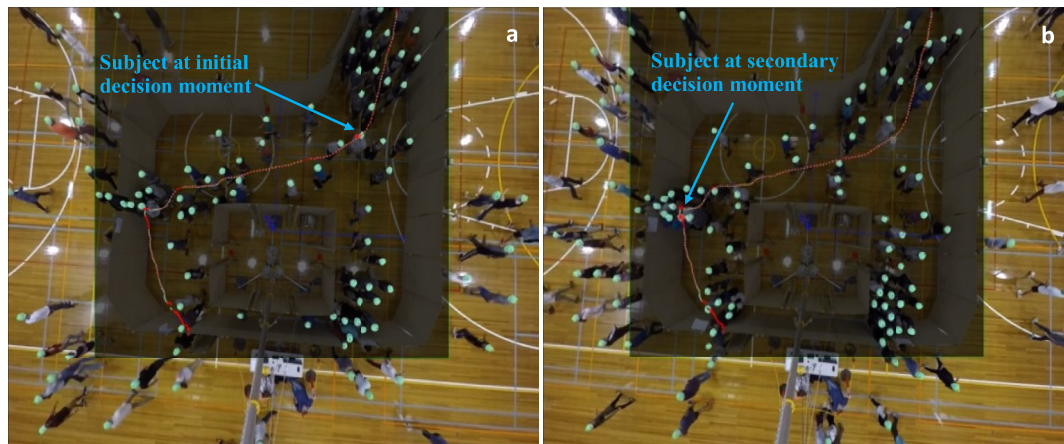


Fig. 6. An example of a participant with more than one decision moment. Figure (a) shows the subject at their initial decision moment and figure (b) shows them at the moment they updated and changed their decision.

coded 0-1 visibility variable, VIS, with 1 representing the visible condition). Individuals were subject to more degrees of attribute perception error compared to the SC experiments in that they had a different perspective than that of the analyst when perceiving the attribute levels (such as the CONG or FLTOEX levels). The analyst investigates each choice situation from a birds-eye view while such perception is not available to the individual who makes the choice inside the crowd. In total 3015 choice observations of this type were extracted by analysing all individuals in all 20 trial runs. The full dataset (of the SC and RC observations) can be accessed in the supplementary materials published online alongside this article (Supplementary Data).

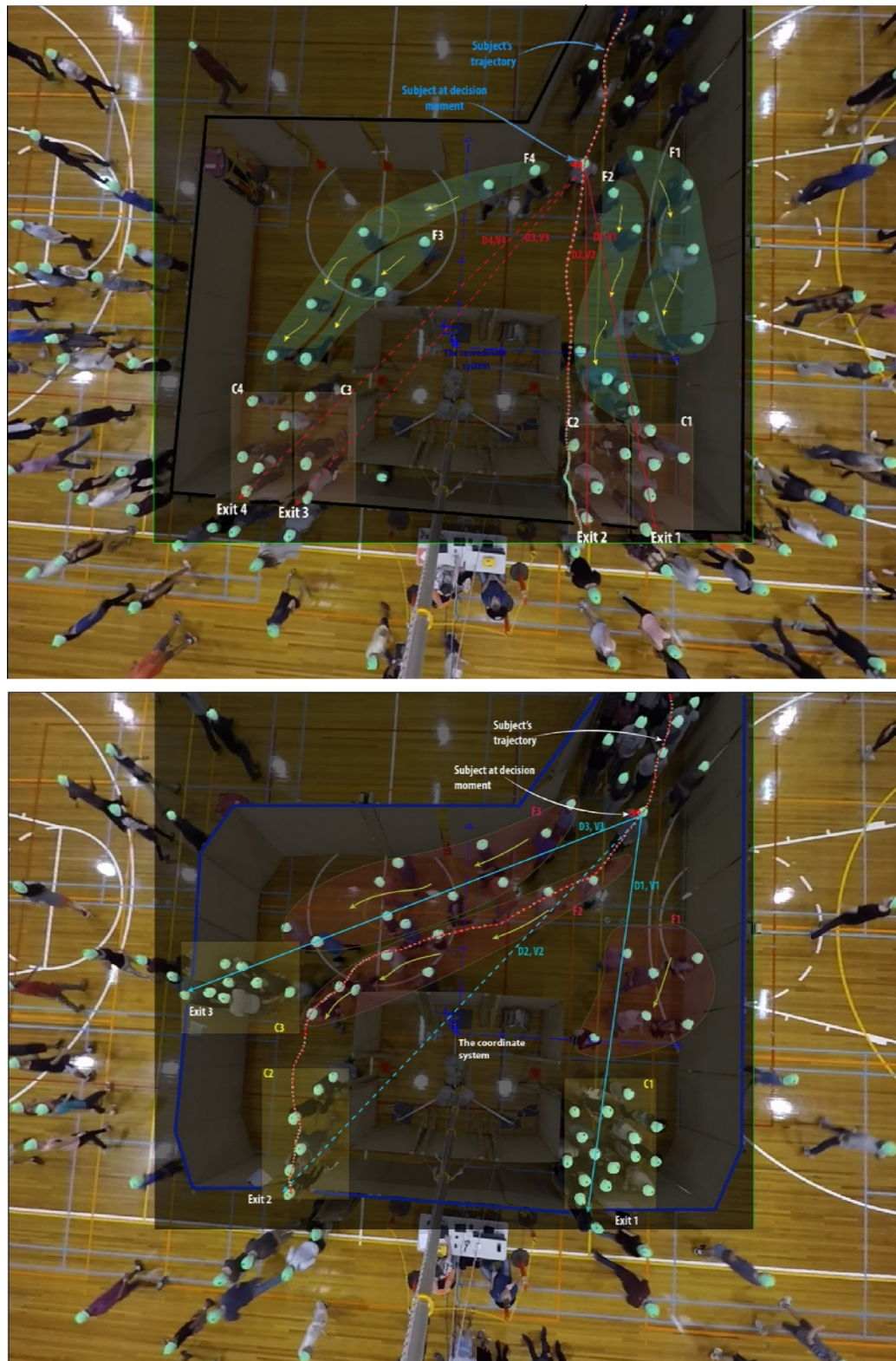


Fig. 7. Extracting choice observations at the identified decision moments. The choice set consists of the available alternatives. The choice (i.e. chosen alternative) can be observed based on the position at which the trajectory has ended. And finally, exit attributes are measured: (C1, C2, ...) represent the size of the crowding near exits, (F1, F1, ...) represent the size of the moving flows, (D1, D2, ...) represent the distance to each exit, and (V1, V2, ...) represent the visibility state of each exit.

Table 1

Mixed logit estimates. The numbers in the brackets show the t -statistic of each estimate (against the test value of 0). Also, the asterisks ***, ** and * indicate the statistical significance at 99%, 95% and 90% level respectively.

Parameter	Estimate SC (I)	SC(II)	SC (III)	RC
Means of utility coefficients				
DIST	−0.306 (−16.88)***	−0.035 (−13.95)***	−0.100 (−5.62)***	−0.277 (−9.08)***
CONG	−0.597 (−17.13)***	−0.181 (−10.33)***	−0.270 (−9.39)***	−0.169 (−10.02)***
FLTOVIS	−0.119 (−7.22)***	−0.015 (−0.36)	−0.233 (−11.42)***	−0.035 (−2.37)**
FLTOINVIS	0.055 (3.04)***	−0.040 (−1.25)	−0.141 (−5.89)***	0.107 (7.75)***
VIS	1.520 (10.06)***	0.509 (2.85)***	1.630 (5.07)***	0.828 (7.49)***
Standard deviations of coefficients				
DIST	0.171 (9.90)***	0.023 (9.63)***	0.002 (0.16)	0.091 (0.93)
CONG	0.327 (11.71)***	0.172 (9.21)***	0.100 (7.44)***	0.181 (4.48)***
FLTOVIS	0.134 (9.27)***	0.259 (8.78)***	0.103 (3.58)***	0.070 (2.92)***
FLTOINVIS	0.136 (9.30)***	0.258 (8.71)***	0.101 (3.47)***	0.072 (2.99)***
VIS	1.190 (6.04)***	1.331 (5.61)***	0.829 (3.80)***	0.091 (0.42)
Statistical fit measures				
Number of choice observations	2338	1410	1210	3015
Initial log-likelihood	−3241.16	−2526.38	−1329.32	−4174.13
Optimum log-likelihood	−2610.42	−1741.50	−1026.44	−2854.74
McFadden ρ^2	0.195	0.311	0.228	0.316
Adjusted ρ^2	0.193	0.309	0.224	0.315

4. Models specification and estimation

4.1. Mixed logit models

Individual n 's choice among J_n alternatives at choice situation t ($t=1, \dots, T_n$), is the one with the maximum utility, where the utility functions are as Eq. 1, consisting of a deterministic part V_{nit} and a random disturbance part ε_{int} . Random unobserved factors are assumed to be distributed identically and independently as (standard/normalized) Extreme Value Type 1 with the density function given as $f(\varepsilon_{int}) = e^{-\varepsilon_{int}} e^{-e^{-\varepsilon_{int}}}$ (see Haghani et al. (2016b) or Haghani et al. (2016a) for further details on the characteristics of this distribution). The vector $\mathbf{x}_{nit}$ is the vector of attributes (Eq. 2), ε_{nit} represents random unobserved factors of the utilities, and β_n is the vector of utility coefficients embodying the relative weights of different factors contributing to the choice of each alternative. The coefficient vector is specified as $\beta_n = \beta + \Gamma \omega_n$ where β is the vector of means of coefficients, ω_n is a vector of independent standard normal variables (with the probability density function denoted as $g(\omega_n)$), and Γ is the Cholesky factor of the covariance matrix, $\Sigma = \Gamma \Gamma'$.

$$U_{nit} = V_{nit} + \varepsilon_{int} = \beta_n' \mathbf{x}_{nit} + \varepsilon_{int} \quad (1)$$

$$\mathbf{x}_{nit} = \left((DIST)_{nit}, (CONG)_{nit}, \underbrace{(VIS)_{nit} \times (FLTOEX)_{nit}}_{=(FLTOVIS)_{nit}}, \underbrace{(1 - VIS)_{nit} \times (FLTOEX)_{nit}}_{=(FLTOINVIS)_{nit}}, (VIS)_{nit} \right) \quad (2)$$

By multiplying the variable representing the size of directional flows (FLTOVIS) by VIS and 1-VIS (i.e. the interaction effect), we basically make a distinction between the marginal effect of the size of the evacuees flows moving to an alternative exit on the exit utility based on the visibility of that exit. When an alternative exit is visible (or equivalently, when the alternative does not present significant levels of attribute ambiguity to the decision maker) the utility coefficient assigned to the size of flow moving to that exit is different to the case in which the exit is invisible to the decision maker (and as a result, the attributes of the exit are partially ambiguous to the decision maker). We refer to these two effects as directional flows to visible and invisible exits being signified as FLTOVIS and FLTOINVIS respectively.

The choice probability of each individual making the same sequence of choices as they were observed to have made in the data $\mathbf{i}_n = \langle i_1, \dots, T_n \rangle$ are given in Eq. 3 and are introduced to the likelihood function as the contribution of individual n to the available information in the data. The estimates are obtained from a simulated maximum likelihood procedure using 1000 Halton draws. The estimate results associated with each dataset for the mixed logit models specified as above are presented in Table 1.

$$P_{in} = \int_{\omega_n} \prod_{t=1}^{T_n} \frac{e^{\beta_n' \mathbf{x}_{nit}}}{\sum_{j=1}^{J_n} e^{\beta_n' \mathbf{x}_{njt}}} g(\omega_n) d\omega_n \quad (3)$$

Table 2

Latent-class logit estimates. The numbers in the brackets show the t -statistic of each estimate (against the test value of 0). Also, the asterisks ***, ** and * indicate the statistical significance at 99%, 95% and 90% level respectively.

Parameter	Estimate SC (I)	SC(II)	SC (III)	RC
Class 1 utility coefficients				
DIST	−0.243 (−16.54)***	−0.017 (−15.38)***	−0.069 (−2.92)**	−0.166 (−4.02)***
CONG	−0.673 (−17.18)***	−0.170 (−14.19)***	−0.250 (−8.68)***	−0.248 (−9.20)***
FLTOVIS	−0.119 (−7.47)***	0.023 (0.87)	−0.259 (−12.27)***	−0.251 (−8.30)**
FLTOINVIS	0.057 (4.02)***	0.001 (0.03)	−0.168 (−7.35)***	0.108 (4.28)***
VIS	1.505 (11.10)***	0.549 (4.66)***	1.400 (4.46)***	1.854 (11.46)***
Class 2 utility coefficients				
DIST	−0.279 (−9.22)***	−0.048 (−12.91)***	−0.110 (−1.67)*	−0.504 (−7.04)***
CONG	−0.178 (−5.19)***	−0.074 (−5.61)***	−0.088 (−0.78)	−0.051 (−1.93)*
FLTOVIS	−0.014 (−0.64)	−0.006 (−0.12)	−0.001 (−0.02)	0.263 (6.68)***
FLTOINVIS	0.087 (2.99)***	0.099 (2.94)***	0.156 (1.85)*	0.069 (2.56)**
VIS	1.057 (3.81)***	0.458 (3.00)***	0.071 (0.24)	−0.132 (−4.06)***
Class probabilities				
Class 1	0.694 (14.75)***	0.606 (11.94)***	0.825 (12.05)***	0.864 (3.02)***
Class 2	0.306 (6.52)***	0.394 (7.78)***	0.175 (2.56)**	0.136 (1.66)*
Statistical fit measures				
Number of choice observations	2338	1410	1210	3015
Initial log-likelihood	−3241.16	−2526.38	−1329.32	−4174.13
Optimum log-likelihood	−2714.11	−1856.77	−1023.85	−2794.37
McFadden ρ^2	0.163	0.265	0.230	0.331
Adjusted ρ^2	0.159	0.261	0.222	0.328

4.2. Latent-class logit models

Individual n 's choice among J_n alternatives at choice situation t ($t = 1, \dots, T_n$), given that individual n is in class c ($c = 1, \dots, C$), is the one with maximum utility with the utility functions are as Eq. 4, where β_c is the vector of class-specific parameters. Given that individual n is in class c , the probability of the individual n choosing the same sequence of alternatives as he/she chose in the sample will follow Eq. 5. The class probabilities are specified by the multinomial logit form (Eq. 6). The class probabilities are simply functions of C parameters, $\theta_1, \dots, \theta_C$, the last of which is normalised to zero due to the model identification considerations. For a given individual, the model's estimate of the probability of a specific choice sequence (the individual n 's contribution to the likelihood function) is the expected value of the class specific probabilities over the classes (Eq. 7).

The process of data analysis strongly suggested the statistical significance of only two categories for all four datasets. Therefore, we estimated models with two classes. The estimation results are presented in Table 2.⁴

$$U_{int|c} = \beta'_c \mathbf{x}_{int} + \varepsilon_{int} \quad (4)$$

$$P_{(i_n|class=c)} = \prod_{t=1}^{T_n} \frac{\exp(\beta'_c \mathbf{x}_{ni,t})}{\sum_{j=1}^{J_n} \exp(\beta'_c \mathbf{x}_{nj,t})} \quad (5)$$

$$P_{(class=c)} = \frac{\exp(\theta_c)}{\sum_{q=1}^Q \exp(\theta_q)}, \theta_C = 0 \quad (6)$$

$$P_{i_n} = \sum_{c=1}^C P_{(class=c)} \times P_{(i_n|class=c)} \quad (7)$$

4.3. Scaled mixed logit model (joint estimation)

The scaled mixed logit model (Fiebig et al., 2009) is estimated on the combined data (SC (I, II, III) and RC data) by assigning specific utility scales to each data segment. The model is specified in the same way as the mixed logit detailed previously, except that the deterministic component of the utilities is defined as Eq. 8, where δ_m 's are context-specific utility scale parameters and z_m 's are dummy-coded 0-1 variables determining the data segment to which each observation

⁴ The extension of the latent-class logit, the random-parameter version, was examined by the authors for all datasets. As in all four cases, the inner-class heterogeneity collapsed into insignificance we simply report here on classical latent-class estimates where the coefficients within each class are assumed constant.

Table 3

The scaled mixed logit estimates. The numbers in the brackets show the *t*-statistic of each estimate (against the test value of 0). Also, the asterisks ***, ** and * indicate the statistical significance at 99%, 95% and 90% level respectively.

Parameter	Estimate
Means of utility coefficients	
DIST	−0.081 (−18.69)***
CONG	−0.175 (−20.27)***
FLTOVIS	−0.062 (−14.40)***
FLTOINVIS	0.031 (6.67)***
VIS	0.807 (16.28)***
Standard deviations of coefficients	
DIST	1.758 (23.25)***
CONG	0.365 (3.15)***
FLTOVIS	0.909 (9.55)***
FLTOINVIS	0.907 (9.54)***
VIS	0.006 (0.01)
Utility scale parameters	
MENZ	0.722 (13.10)***
FLIND	−1.571 (−26.56)***
PARL	0.211 (2.97)***
Statistical fit measures	
Number of choice observations	7973
Initial log-likelihood	−14,278.53
Optimum log-likelihood	−9051.00
McFadden ρ^2	0.366
Adjusted ρ^2	0.365

belongs. The scale parameter associated with the RC data sector is normalised at 0 for the model identification purposes. Table 3 shows the estimation results for this model.

$$V_{nit} = \exp \left(\sum_{m=1}^3 \delta_m z_m \right) \beta'_n x_{nit} \quad (8)$$

5. Model comparisons and interpretations

5.1. Models estimates

We contrast in this section the parameter estimates derived from all four data sources reported in the previously. In order to facilitate the comparison, we visualise the 95% confidence intervals of the estimates for the mean of utility coefficients of the four mixed logit models in Fig. 8 and those of the class-specific utility coefficients of the latent-class models in Fig. 9. The sign and statistical significance (at 95% confidence level) of the estimates can as well be identified from the figures.

5.1.1. Mixed logit estimates

According to Fig. 8, there is full consensus between all four observational sources on the central tendency of the estimated mixed logit parameters in terms of the sign and significance of three variables including distance (DIST), congestion (CONG) and visibility (VIS). All four models agree that the three variables contribute significantly to the exit choices of evacuees and that the direction of influence on the exit utility for the first two is negative as opposed to the positive marginal utility impact of the visibility variable. In other words, it is suggested by all four data sources that the further an exit is located or the heavier the congestion around the exit area, the less likely the decision maker is to choose that exit as the quickest escape option, a finding that was also intuitively expected. Also as expected, the estimate for the dummy-coded variable (i.e. VIS) is associated with the highest level of uncertainty (i.e. the greatest standard error) which is a typical feature of the categorical variables in discrete choice models.

However, in relation with the magnitude of the estimates associated with the DIST and CONG variables relative to one another, it is interesting to note that the estimates inferred from the field-type realistic context tells a different narration compared to all three sets of SC observations. In all three SC-generated models, the confidence interval associated with the CONG variable locates at a lower position than that of the DIST variable, whereas this pattern is reversed with the realistic context. In other words, the SC-generated contexts suggest a stronger marginal effect for the CONG variable “relative to” the DIST variable, whereas the experiments collected in the realistic context suggest the contrary. The fact that all three SC-generated models are consistent in this aspect of their estimates and that does not match with what observed in the realistic choices suggests that this could be due to a systematic bias in the perception of these two variables relative to one another

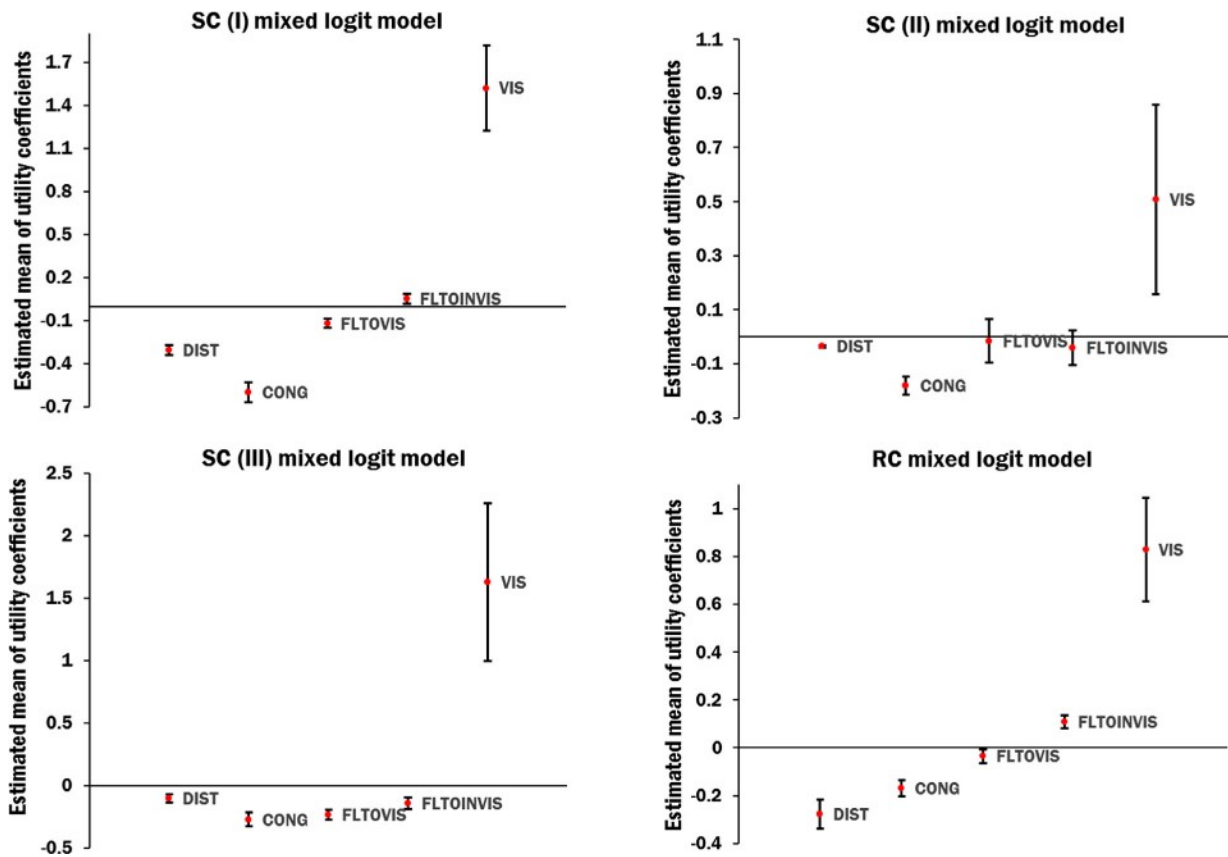


Fig. 8. Mixed logit estimates obtained from the four data sources. The 95% confidence intervals of the mean of utility coefficients are visualised along with the estimates of the mean.

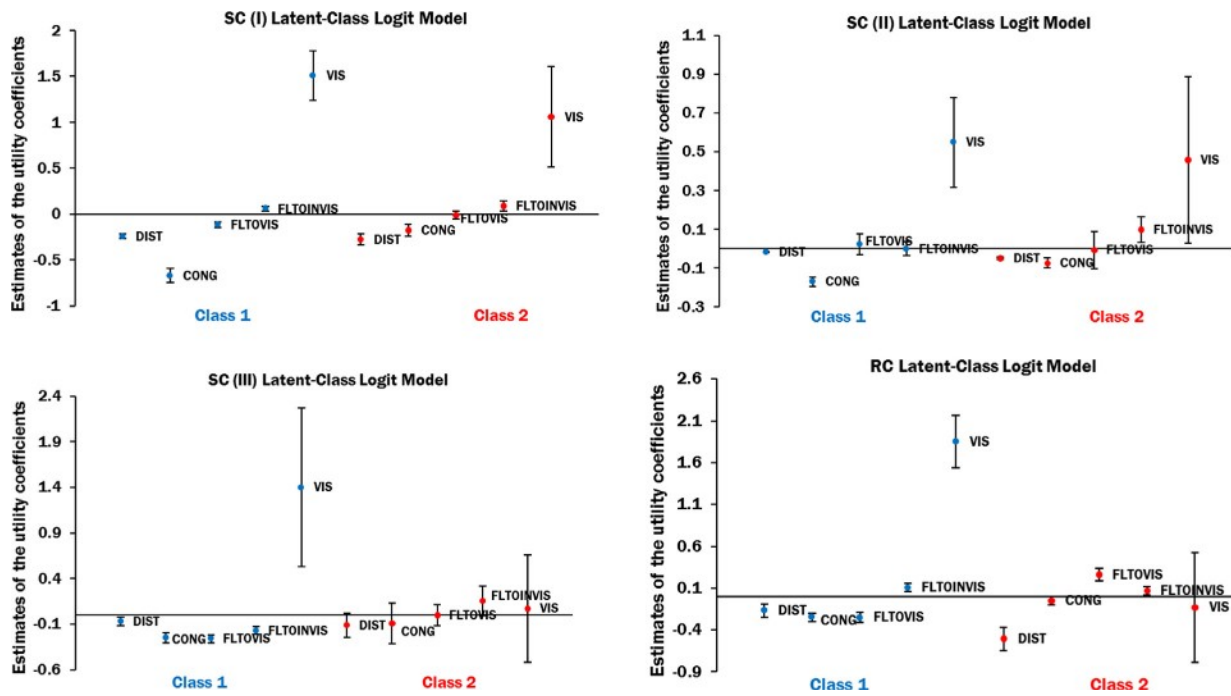


Fig. 9. Latent-class logit estimates obtained from the four data sources. The 95% confidence intervals of the class-specific utility coefficients are visualised along with the estimates of the coefficients.

when introduced on a picture in contrast to the realistic decision situations. It appears that when facing the hypothetical choice situations presented as pictures, individuals tend to underestimate the effect of minimising spatial distances on their preferences relative to avoiding heavily congested areas compared to the situation where they have to make this decision in real actions (and as a result, traverse the distance by actually reaching the chosen exit). In other words, when facing a choice situation as a picture, the impact of congestion becomes more prominent (i.e. the saliency effect) relative to choosing the closest exit in contrast to the decisions that they make in real actions. However, as a referee pertinently pointed out, this could be just one possible explanation and we cannot rely heavily on this interpretation considering that we did not (and in fact, cannot) test for it. Therefore, while one might see this as a plausible explanation, we tend to remain cautious about the possible over-interpretation of this observation.

For the variables related to the effect of directional flows, that is to say FLTOVIS and FLTOINVIS, we did have strong prior expectation about the sign and the direction of marginal influence on the utilities and that was to be explored in the modelling process. Three of the data sources (i.e. the SC (I), SC (III) and RC) suggest the effect of the directional flows in either form (whether moving to visible or invisible exits) significant to the choice of exit. They all also agree on the direction of impact associated to this variable on the utility of visible exits (i.e. FLTOVIS), as a negative utility element. They all suggest that when the decision maker can see an exit and is thus aware of the attributes of that particular exit, the movement of a large number of evacuees towards that particular alternative do not increase the desirability of that alternative; rather it reduces the probability of that alternative to be chosen. In other words, individuals tend to perceive such flows as a form of (potential) congestion (or further delay) with a more moderate marginal effect compared to the actual jamming around exits (i.e. CONG). However, it is suggested by these models that the marginal effect of the directional flows on an exit utility differs when the chosen target of the flow is not visible to the choice maker. Once facing (partial) attribute ambiguity associated with an alternative exit (caused by the invisibility of the exit) individuals tend to perceive a less negative (or even positive) effect from observing a large number of evacuees choosing that alternative. The estimates obtained from the SC (I) and RC sets suggest positive marginal effects for FLTOINVIS variable.

As to the reason why both FLTOEX variables were estimated as insignificant from the SC (II) survey our possible explanations is that this could be attributed to the considerable difference between the size of the venue to which SC (II) survey referred in contrast to the three other experiment locations. The train station to which the choice scenarios of the SC (II) survey referred is by far larger than the three other locations in the three other choice contexts. As a result, the magnitudes of the spatial distance (as well as their differences) in the SC (II) data are far larger than those of other choice contexts in our observations. It appears that in such situations, the decisions of the individuals are dominantly influenced by the DIST, CONG and VIS and the movement of the other people towards different directions does not significantly impact on their choice. In other words, for example, the difference between the distances to different exits could be so large in such venues that the “cost” of giving up on the closest exit and instead following a group of evacuees headed to another alternative could be unreasonably large for the decision maker to be even considered. It appears that in venues as large as a major train station, individuals may not perceive the positive or negative impact of the moving flows as significantly as observed in the other locations. This in part could be due to fact the drop in the capacities arisen from of the movement of other individuals to different directions might not be perceived consequential by decision makers when the capacity of the venue is already very large. Yet, we suggest this only as a possible explanation as we one should clearly be cautious about making definite conclusions from insignificant estimates, and accordingly we suggest that this explanation needs to be further investigated.

5.1.2. Latent-class logit estimates

The major common aspect of the modelling emerged from our latent-class logit analyses upon which all four data sources clearly and strongly agreed was the relevance of only two behavioural classes of evacuees. According to Fig. 9, in which the coefficient estimates associated with each class has been visualised, it can be observed that each model suggests a class residing in the majority that display fairly similar behavioural patterns to what observed with the corresponding mixed logit estimates (the population split suggested by each model is given in Table 2). According to the estimates, the major difference between the two classes can be attributed to the perception of the social influence factors (i.e. CONG, FLTOVIS and FLTOINVIS) relative to the DIST. It is suggested by the models that the minority class perceives less disutility from the presence (i.e. CONG) and the movement (FLTOVIS and FLTOINVIS) of other evacuees relative to the DIST factor in comparison with the perception of the majority class who place relatively higher priorities on avoiding other individuals when making their choice of exit. In contrast, the minority class displays stronger propensity to assign greater relative weights on choosing the closest exits and tend not to care as much about avoiding heavy congestions as individuals from class 1 do.

5.1.3. Scaled mixed logit estimates

A joint estimation on the combined data allowed us to investigate the potential differences in the scale of the utility estimates associated with each of the four observation sets. By disentangling the context-specific scale difference of the utility estimates one would be able to differentiate between the randomness degrees of the decisions made in the SC contexts relative to the realistic setting. According to the results presented in Table 3, the estimates associated with two of the SC data segments are of a greater scale relative to the observations made in the field-type context. This includes the SC (I) and SC (III) datasets, whereas the observations in the SC (II) data result in estimates of a smaller scale than that of the RC context. Therefore, it appears that decisions made in the real choice context are likelier to be made more randomly than what

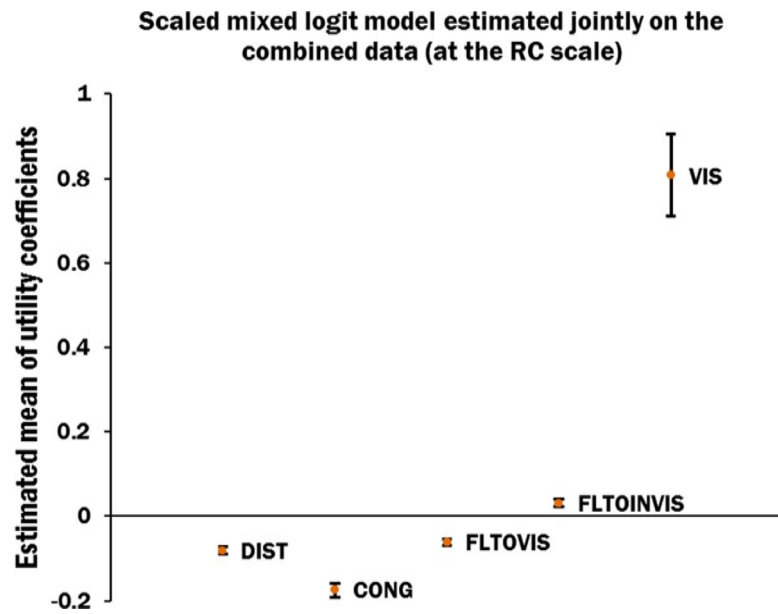


Fig. 10. Scaled mixed logit estimates obtained from the combined data. The 95% confidence intervals of the mean of the utility coefficients are visualised along with the estimates of the mean at the scale associated with the RC data segment.

people decide in a hypothetical context where they have a clearer perception of the different attributes. Also, specific to this particular context of choice, individuals have significantly less amount of time to make a choice in the realistic context as they move and escape in a real crowd compared to the hypothetical settings in which they can ponder upon their choice and evaluate different attribute levels more effectively. Also, when in an actual crowd they have a less accurate perception of the different attributes relative to the top-down view practised in our SC contexts. For the sake of comparison with the model estimates derived from individual datasets, we have also visualised in Fig. 10 the confidence intervals of the estimates obtained from the combined data, at the scale corresponding with the RC segment of the pooled data.

5.2. Models predictions

In the previous sections, we elaborated on the similarities and dissimilarities of the utility estimates derived from the stated and realistic choice contexts. We found similarities in terms of the estimate patterns particularly on the sign and significance of the coefficients. We also detected dissimilarities notably in terms of the scale of the utility parameters. Here, we investigate the extent to which these similarities and dissimilarities might overall impact on the predictions of the models. We would like to quantify the extent to which our predictions would have been affected if we only had collected only a single dataset using the SC methods compared to the prediction accuracy achieved by conducting the field-type experiments. In doing so, we contrast the predictions of the three SC mixed logit models to those of the RC model as well as to those of the model estimated on the combined data. To this end, by removing the choice (i.e. the dependant variable) element from the 3015 choice observations, we set them as 3015 “choice situation” upon which we base our comparisons.

In Fig. 11, we contrast the probabilities predicted by each of the three SC datasets with those of the model inferred from the realistic experiments. In each scatterplots shown in Fig. 11, each point corresponds to a particular alternative in one of the 3015 choice situations described earlier. The coordinates of each point are the probabilities generated by an SC model (represented by vertical axis) and the corresponding probabilities generated by the RC model (represented by the horizontal axis). The bisector line $y=x$ is also displayed to facilitate the comparison. Clearly, the less dispersedly the points are scattered around this line, the more similar the predictions of the two models are. Corresponding to each scatterplot, we provide a graph in which the concentration of the points on the plane of coordinates (i.e. the number of points per unit square) is colour-coded. The absolute difference between the corresponding probabilities are also calculated, whose distributions are shown against the corresponding scatterplots. The distribution of these pairwise differences of predicted probabilities are also visualised in a different form in the box-and-whisker plots presented in Fig. 12. The figure also presents the average of these pairwise differences (calculated on the basis of the absolute values). Similar analyses are conducted by comparing the prediction of the models obtained from each dataset, SC (I, II and III) and RC, to those of the jointly estimated model, and the corresponding results are presented in Figs. 13 and 14.

According to these analyses, it turns out that each single dataset of the stated choices produces choice probabilities that are fairly similar to the prediction of the model inferred from the RC context (or alternatively, the predictions of the joint model). According to Fig. 11, predictions of the SC (I, II and III) models show a strong pattern of correlation with those of

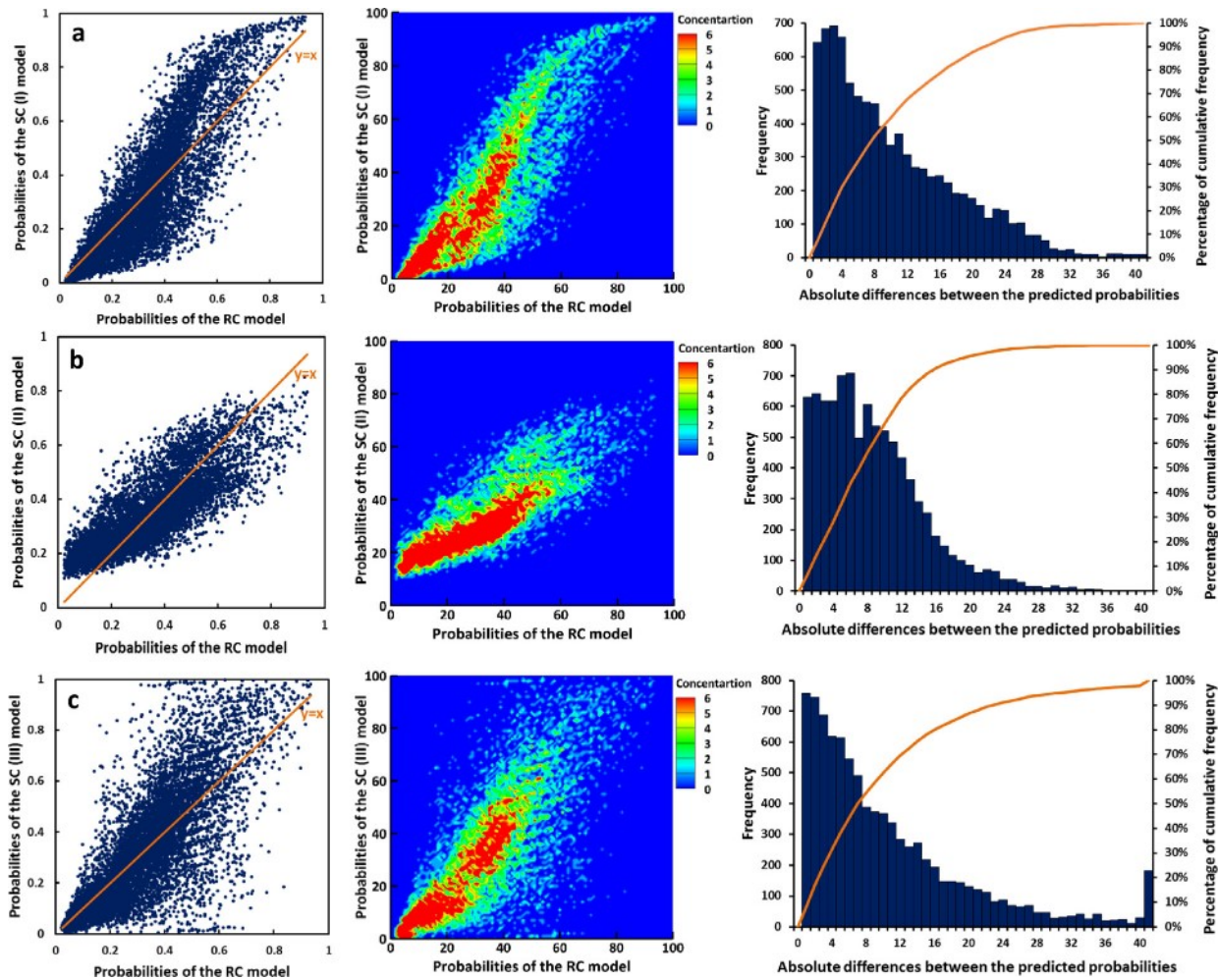


Fig. 11. Comparing the predictions of the models inferred from the SC experiments to those of the model derived from the realistic field-type (i.e. RC) experiment. Figure series (a), (b) and (c) (aligned horizontally) respectively compare the SC (I), SC (II) and SC (III) models to the RC model. The coordinates of each point in the scatterplots present the probabilities predicted by the models represented by the two axes. The dispersion of the points around the line $y = x$ is a measure of (dis)similarity between the predictions of the two models. The concentration of these points is colour-coded in the graphs next to each scatterplot. The histograms show the distribution of the (absolute) differences between the corresponding probabilities of each pair of the models represented by each of the three figure series.

the RC model. The correlations are 0.86, 0.82 and 0.78 respectively, while the absolute differences between their probabilities average at 9.76%, 8.06% and 10.11% respectively. In other words, each of the models derived from the SC experiments produce probabilities that are on average no more than 10% different from the probabilities of the RC model. According to the histograms of the distributions of the absolute differences in Fig. 11, the frequency of the observations rather rapidly decreases as the value of the absolute difference between the model probabilities increase (i.e. as one moves along the x-axis) with most observations distributed around low values of the absolute differences. We calculated the 85th percentile associated with each dataset, that is to say the percentage of the absolute differences (i.e. the x-axis value) below which are 85% of the observations. The calculated figures are 19%, 14% and 19% respectively for SC (I), SC (II) and SC (III) datasets. In other words, for 85% of the probabilities generated by the SC (I) model, the difference with the corresponding RC probability is less than 19%. Similar interpretation can be applied to SC (II) and SC (III) given the abovementioned values.

Also according to Figs. 13 and 14, the predictions of each SC model on average differs from the predictions of the joint model no more than 9%. The average differences associated with the SC (I–III) and RC are respectively 8.46%, 5.18%, 8.86% and 5.57% with their predictions also being correlated at 0.94, 0.92, 0.88 and 0.90 levels with the probabilities of the joint model. According to the histograms of frequencies in Fig. 13, the corresponding 85th percentiles are respectively 15%, 9%, 15% and 10%. Also, according to Fig. 13, the RC generated predictions show the least amount of systematic overestimation and underestimation of the probabilities generated by the joint model as the probability scatter is more symmetrical around the line $y = x$.

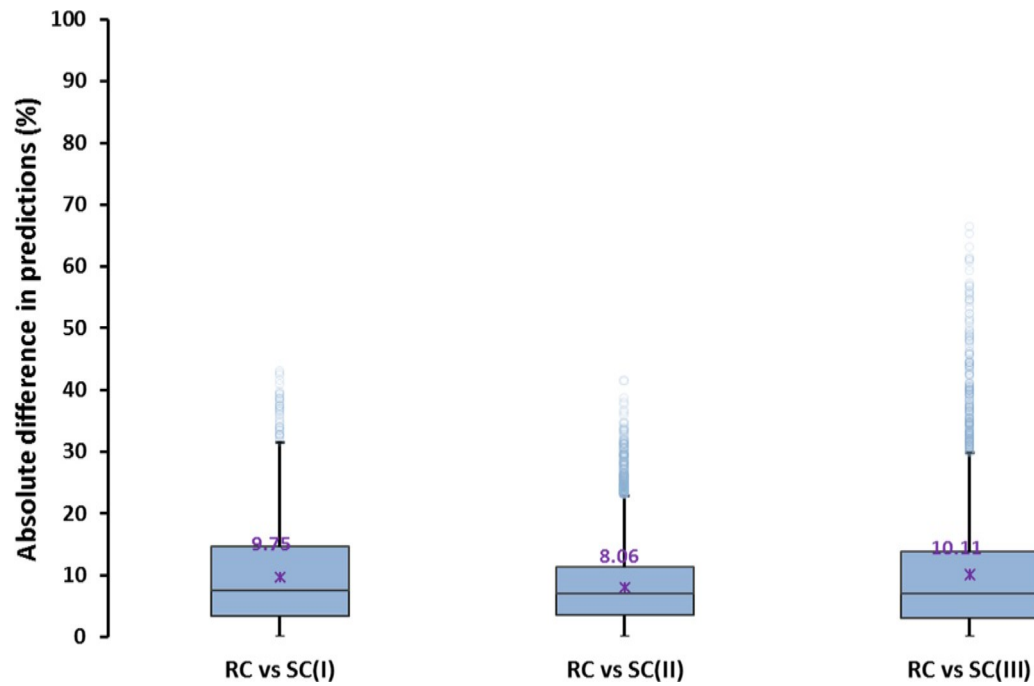


Fig. 12. Comparing the predictions of the models inferred from the SC experiments to those of the model derived from the realistic field-type (i.e. RC) experiment. The absolute pairwise differences between the corresponding probabilities generated by each pair of the models were calculated and the box-and-whisker plots show the distribution of these differences. The outliers (defined as the observations “greater than the third quartile plus 1.5 times the interquartile range” are presented individually. The asterisk marks represent the mean of the differences between the predictions of each model pair.

Considering the substantial difference between the contexts from which these predictions have been derived (i.e. looking at pictures presenting primitive scenarios and making decisions as opposed to making decisions in real actions) the similarities are rather surprising and unexpected. Considering the level of accuracy that is actually needed in practice for modelling evacuation scenarios, an average of 10% error for instance might not be excessively intolerable in many substantive cases. A great range of uncertainties are in place which makes it rather impractical to model every single decision of individual evacuee with precision. However, as long as the aggregate patterns of the movement (such as densities, evacuation time etc.) can be replicated with adequate precision, one would be able to obtain satisfactory results from the models and rely comfortably on the resultant policies.

This offers great promise as to the applicability of calibration efforts based on SC methods in this context owing to the fact that the RC data (whether experimental or real-world) is often unavailable to the analysts. There appears to be considerable elements of truth to such observations, more than the extent that they can be ruled out as a fully invalid method. In other words, the results suggest that just because the choices have been made in hypothetical scenarios does not mean that they cannot be used for practical applications. It even offers greater promise as to the relevance of investigating the behaviour in immersive virtual reality environments that hypothesise evacuation scenario as a close representation of the actual environments. One would expect to achieve even closer predictions to those of the realistic choice if the scenarios had been framed in the form of three dimensional representations of the surveyed environments. This would enable the analyst to benefit from reliable data and calibrate their model at reasonable amounts of time and cost without posing any danger to the participants. As opposed to using arbitrarily-specified parameter values that might not be adequate reflections of the behaviour and can result in misled policies, this could certainly be regarded as a practical solution.

6. Summary and conclusions

We addressed the particular problem of the context dependency and sample-to-sample variations of the modelling results when emergency escape behaviour of pedestrian evacuees is derived from their choices in virtual decision environments. In doing so, we reported on the choice observations obtained from three SC surveys of emergency exit in which the choice experiments presented a reference to the actual geometry of the place in which the experiment was carried out. For the first time on this topic, we also derived a fairly large set of realistic choice (i.e. RC) observations of emergency exit through staging a series of simulated escape experiments with actual crowds, and observing the decisions of individuals in realistic actions. Our main aim was to identify the similarities and dissimilarities of the behavioural findings derived from each data source as well as investigating the extent to which their predictions differ from one another.

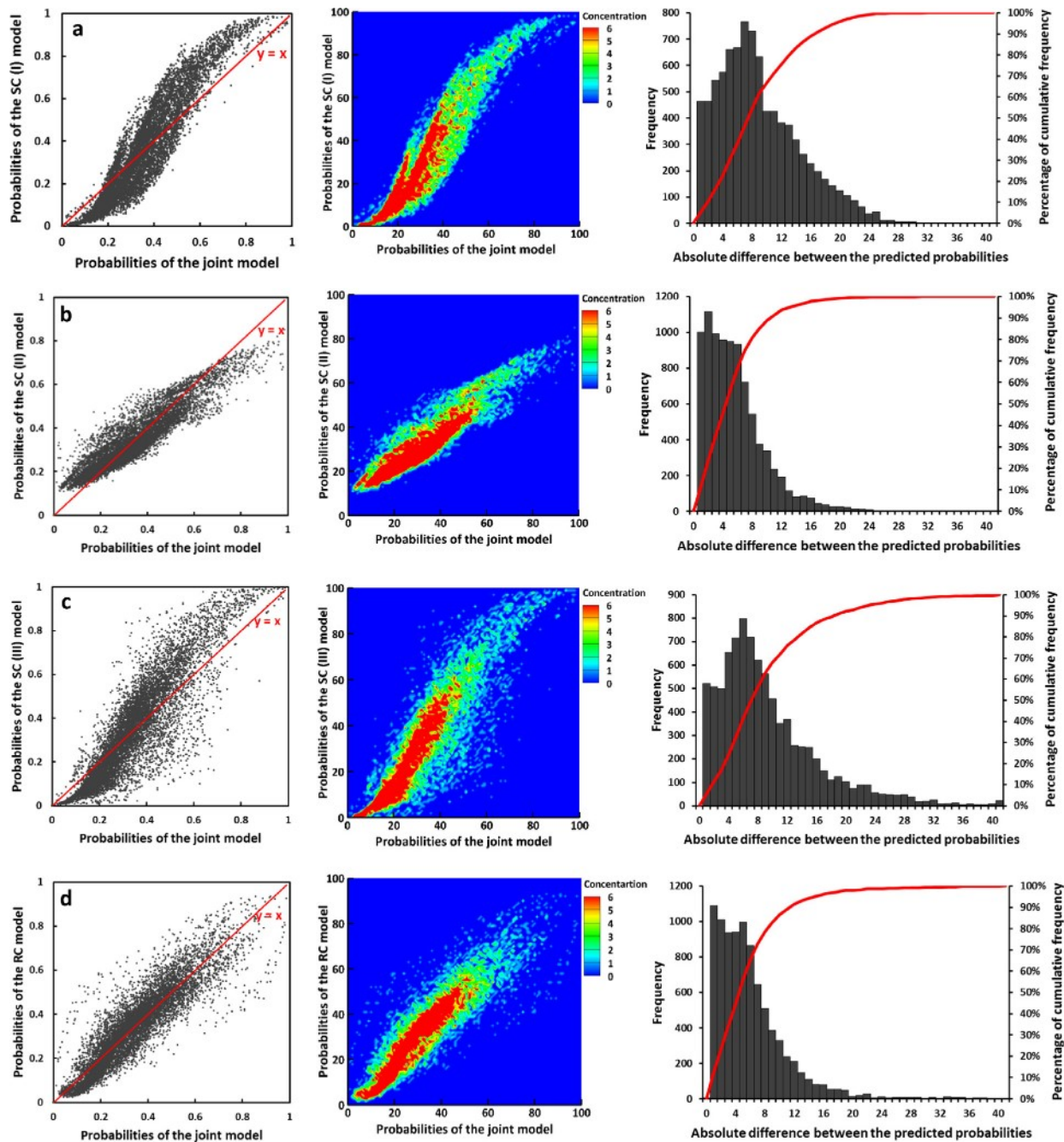


Fig. 13. Comparing the predictions of the models inferred from the SC experiments as well as the RC experiment to those of the model derived from the pooled dataset (at its RC scale). Figure series (a), (b), (c) and (d) (aligned horizontally) respectively compare the SC (I), SC (II), SC (III) and RC models to the joint model. The coordinates of each point in the scatterplots present the probabilities predicted by the models represented by the two axes. The dispersion of the points around the line $y = x$ is a measure of (dis)similarity between the predictions of the two models. The concentration of these points is colour-coded in the graphs next to each scatterplot. The histograms show the distribution of the (absolute) differences between the corresponding probabilities of each model pair.

Our analyses revealed surprising levels of consistency between the model estimates patterns inferred from each data source regardless of the context of their origin. All four data sources consistently agreed on a number of major behavioural elements corroborating the suggestions of one another on those behavioural aspects. It was to the agreement of all derived models that the factors of spatial distance, congestion level, exit visibility and the size of moving flows all significantly contribute to the exit choices of evacuees for a wide range of geometry types and venues. However, for very large venues at the scale of major transport stations, the movements of other evacuees towards different escape alternatives did not appear

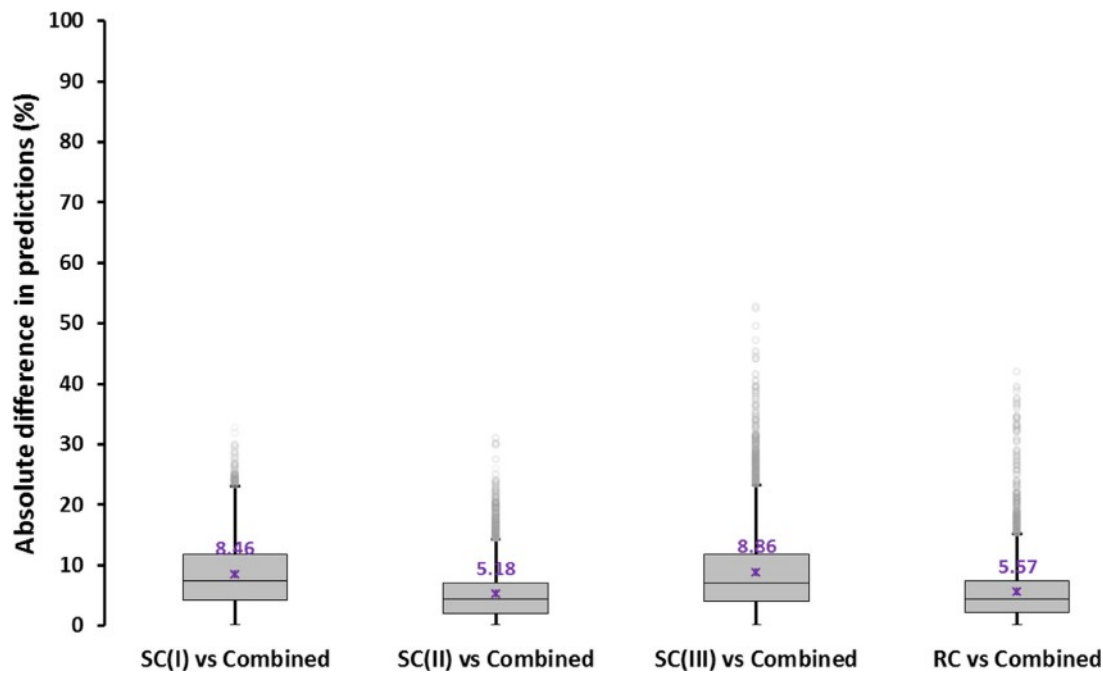


Fig. 14. Comparing the predictions of the models inferred from the SC experiments as well as the RC experiment to those of the model derived from the pooled dataset (at its RC scale). The absolute pairwise differences between the corresponding probabilities generated by each pair of models were calculated and the box-and-whisker plots show the distribution of these differences. The outliers (defined as the observations “greater than the third quartile plus 1.5 times the interquartile range” are presented individually. The asterisk marks represent the mean of the differences between the predictions of each model pair.

to be statistically significant. There appears to be certain levels of dependency between the behavioural findings emerged from the SC methods in this topic and the particular features of the environment visualised in the choice scenarios. It was also suggested by our models that the perception of observing the exit decisions of other evacuees can significantly differ depending on whether or not the attributes of their chosen target is fully known to the decision maker. It was also suggested by our datasets that the perception heterogeneity of evacuees in terms of their exit choices is classifiable to two major behavioural types, whose primary difference lies within the perception of the congestion-related (i.e. social influence) factors (i.e. the presence and movements of other evacuees), with the majority being more prone to base their decisions primarily based on avoiding heavy congestions.

Despite the major elements of consistency between the estimates derived from all four sources of observations, we also observed certain extents of systematic context-dependent variations that could be taken into consideration when using the SC methods in this particular choice topic. Most importantly, there appeared to be a systematic pattern of perception bias (in the form of underestimation) in relation with the spatial distances when scenarios are visualised as a top-down picture. Also the jointly estimated model on the pooled data suggested that the randomness degree of the decisions can be significantly underestimated by this method relative to what was actually observed in the realistic setting.

Nevertheless, in terms of the predictions, the models originated from each sources of SC proved to generate probabilities that were not dramatically different to those of the RC-generated model. With fair level of precision, we could have made by-and-large same predictions using each of the single SC sources as we did using the RC model or the model inferred from the combined data. This provides promising evidence as to the potential applicability and the relevance of the behavioural findings or the model predictions derived from the SC experiments on this topic even in its most modest forms, particularly when data from naturally-occurring or realistic settings are prohibitively unavailable. This could as a result have major implications for the continuation of research on this particular topic according to the scarcity of the real-world evacuation data in the form of modelling material.

As a further step towards the particular problem addressed in our study, we suggest reinvestigating this question by looking at the extent to which improving the realism level of virtual visualisation may impact on the accuracy of the resultant choice data. This could be majorly important according to the great experimental design possibilities that the recent developments of the immersive virtual-reality technology can potentially offer to the researchers in this area (Moussaïd et al., 2016; Ronchi et al., 2015; Tang et al., 2009). Most interestingly, as this paper was under review, the study of Moussaïd et al. (2016) emerged where the researchers utilised a shared three-dimensional virtual environment as a close proxy of the real crowd evacuations. It is interesting to note how the evidence offered by their independent (and to large degrees, different) experiment suggest similar insight to that of this paper, substantiating our main message: that there is great potential

for learning from suitably-designed virtual evacuation experiments; and that for eliciting people's behaviour we might not necessarily need to observe them in real actions. The authors will report in subsequent works on evidence as to the possible improvements that can be achieved through a more realistic representations of the environment using a three dimensional model of one of the train stations that was used as a reference in their surveys. Particularly for investigating the aspects of behaviour in which the decision-making is of main interest (rather than the local dynamics of the crowd) this could provide a major step forward and a pragmatic solution for dealing with the lack of reliable empirical data in this context.

Also considering that here in this work we used an axiomatic approach of modelling choices (i.e. the utility maximisation) an important question would be whether the use of a different axiomatic paradigm would lead to different behavioural findings. At least one alternative approach is to assume evacuees as regret minimisers instead. We suggest further investigations of the implications of the utility maximisation assumption in this context. The authors may also report on regret minimisation analyses of these observations as an attempt towards examining the aforesaid question.

Also the set of parameters used in our survey design as well as the experiment analysis is by no means claimed to be a comprehensive set of contributing factors to the exit choice. This was mainly restricted by the practical considerations of the experiment design that do not readily allow one to accommodate and estimate more than typically 4 to 5 parameters base on a reasonable number of observations. The authors conducted a scoring pilot survey prior to the survey design and the chosen variables were scored highly but so did some other factors such as the familiarity of the exits. Exploring the role of these factors whose effect had to be downplayed by our study limitations could provide topics for further research on this topic. Particularly, how the behaviour of evacuees changes when individuals are largely unfamiliar with their surrounding environments still remain as a relatively unexplored topic.

Our focus in this work was primarily on comparing the parameter estimate as well as the prediction patterns generated by the SC and RC models. An alternative approach for interrogating the validity of these models would however be integrating the proposed exit choice model with an operational-level crowd simulation method and examine how the movement patterns generated by the combined model compare to the observed trajectory patterns. As the models reported in this work are all fully generic in parameters, implementation would be relatively straightforward through which one can use different sets of parameters suggested by our study and see the impact based on the aggregate (or even disaggregate) measures of behaviour. The authors will in following works report on such analyses while at the same time we also suggest independent investigations of the parameter sets proposed by this work based on independent datasets.

Also, question might be arisen as to which parameter sets to use if one is to choose between the different models proposed in this work. Our answer, according to our analysis, would be that it probably will not matter so much as all parameter sets were turned out to be generating by-and-large similar probabilities. But on the other hand, it would be preferable to use the parameters obtained from the joint estimation on the combined data (Table 3) which benefit from a larger dataset (nearly 8000 observations) and a greater range of variability in terms of the attribute levels. As to which scale to choose, one would recommend the scale associated to the RC segment of our data as it has been driven from a more natural setting. The raw values provided in Table 3 are already at the RC scale.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.trb.2016.10.019](https://doi.org/10.1016/j.trb.2016.10.019).

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